Experiment 10

Question

Perform Moving (Translation) of an Image from One Place to Another.

Aim

To move (translate) an image from one position to another using Python and OpenCV.

Procedure

1. Import the OpenCV (cv2) and NumPy (numpy) libraries.
2. Read the image using cv2.imread().
3. Check whether the image is loaded successfully.
4. Display the original image.
5. Create a translation matrix.
6. Apply the translation using cv2.warpAffine().
7. Display the translated image.
8. Save the translated image.

Code

```
import cv2

import numpy as np


# Read image

image = cv2.imread("penguin.jpg")
```

```
# Check whether image is loaded
if image is None:
    print("Image not found!")
else:
    # Display original image
    cv2.imshow("Original Image", image)

    # Get image size
    rows, cols = image.shape[:2]

    # Translation matrix
    M = np.float32([[1, 0, 100],
                    [0, 1, 50]])

    # Move the image
    moved = cv2.warpAffine(image, M, (cols, rows))

    # Display translated image
    cv2.imshow("Moved Image", moved)

    # Save output image
    cv2.imwrite("moved_penguin.jpg", moved)
```

Input

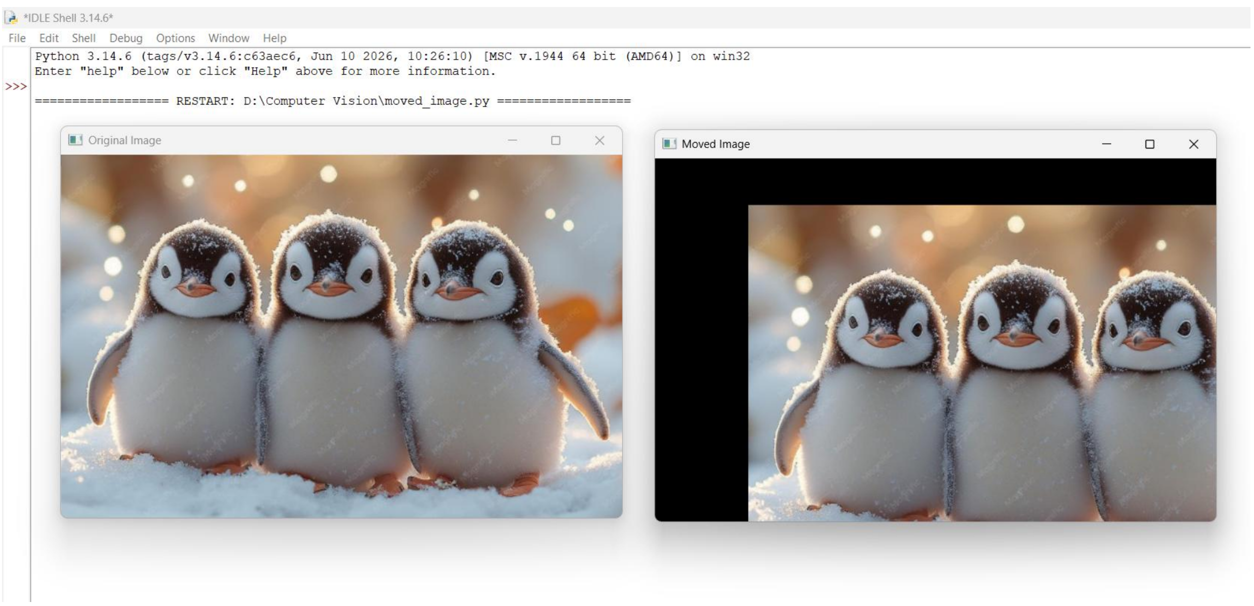
Input Image: penguin.jpg

Output

1. Original Image
2. Moved (Translated) Image

Saved file:

- moved_penguin.jpg



Result

The image was successfully translated (moved) from its original position using OpenCV. The translated image was displayed and saved successfully.